

Hyunju Kim

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RESEARCH INTERESTS

Computational Neuroscience

Apply deep learning to understand neural dynamics, brain connectivity, and functional representations. Build brain-inspired deep learning models based on neural processing principles.

Graph Data Mining & Graph Algorithms

Design algorithms and AI models for analyzing large-scale, complex graph data.

EDUCATION

Georgia Institute of Technology

Ph.D. Student in Computational Science & Engineering. Advisor: Nabil Imam

Aug. 2024 –

Atlanta, GA, United States

KAIST (Korea Advanced Institute of Science and Technology)

M.S. in Artificial Intelligence. Advisor: Kijung Shin

Feb. 2022 – Feb. 2024

Daejeon, Republic of Korea

POSTECH (Pohang University of Science and Technology)

B.S. in Mathematics. Summa cum laude. GPA 3.96/4.3, Major GPA 4.02/4.3 (ranked 1st)

Feb. 2018 – Feb. 2022

Pohang, Republic of Korea

PUBLICATIONS

*: equal contributions C: Conference J: Journal P: Preprint

- [P3] Spatiotemporal Hyperedges for EEG Seizure Detection and Prediction. *Under review*. **Hyunju Kim**, SheoYon Jhin, Noseong Park, Nabil Imam
- [P2] Geometric Learning for Predicting Functional Connectivity on Folded Cortical Surfaces. *Under review*. **Hyunju Kim**, Michael Arcaro, Nabil Imam
- [P1] Geometric Constraints in the Development of Primate Extrastriate Visual Cortex. *Under review*. **Hyunju Kim**, Michael Arcaro, Nabil Imam (DOI: 10.64898/2026.02.04.703881)
- [J3] Four-Set Hypergraphlets for Characterization of Directed Hypergraphs. *TKDE*. Heechan Moon*, **Hyunju Kim***, Sunwoo Kim and Kijung Shin (DOI: 10.1109/TKDE.2026.3677137)
- [J2] Estimating Simplex Counts via Sampling. *The VLDB Journal*. **Hyunju Kim***, Heechan Moon*, Sunwoo Kim and Kijung Shin (DOI: 10.1007/s00778-024-00890-9)
- [J1] Hypergraph Motifs and their Extensions beyond Binary. *The VLDB Journal*. Geon Lee*, Seokbum Yoon*, Jihoon Ko, **Hyunju Kim** and Kijung Shin (DOI: 10.1007/s00778-023-00827-8)
- [C3] DiffIM: Differentiable Influence Minimization with Surrogate Modeling and Continuous Relaxation. *AAAI 2025*. Junghun Lee, **Hyunju Kim**, Fanchen Bu, Jihoon Ko and Kijung Shin
- [C2] FlowerFormer: Empowering Neural Architecture Encoding using a Flow-aware Graph Transformer. *CVPR 2024*. Dongyeong Hwang, **Hyunju Kim**, Sunwoo Kim and Kijung Shin (DOI: 10.1109/CVPR52733.2024.00586)
- [C1] Characterization of Simplicial Complexes by Counting Simplexes beyond Four Nodes. *WWW 2023*. **Hyunju Kim**, Jihoon Ko, Fanchen Bu and Kijung Shin (DOI: 10.1145/3543507.3583332)

RESEARCH EXPERIENCES

EEG Analysis for Cognitive and Clinical Modeling

Sep. 2024 –

Georgia Tech

Drift-Diffusion Model (DDM) analysis of EEG signals for characterizing decision-making processes. EEG-based disease prediction leveraging neural biomarkers and deep learning methods.

fMRI-driven Graph Modeling of Self-Organization Mechanisms in the Visual Cortex

Sep. 2024 –

Georgia Tech

Develop graph generation methods using both AI-driven approaches and rule-based algorithms. Design brain-inspired models by leveraging neural data visualization mechanisms [P1, P2].

KAIST

Designed scalable algorithms for counting and characterizing higher-order structures (simplets, hypergraphlets, motifs) in simplicial complexes, hypergraphs, and directed hypergraphs.
Developed sampling-based estimation methods with provable accuracy guarantees [C1, J1, J2, J3].

Development of a Null Model for GNN Incorporating Node Features

May 2023 – Dec. 2023

KAIST

Developed a null model maintaining performance rankings aligned with the original network while anonymizing users’ information.

Performance Prediction of Architectures using Semi-Supervised Learning

Apr. 2022 – Dec. 2023

KAIST

Neural architecture performance prediction method using a flow-aware Graph Transformer.
Registered two South Korea Copyrights for Software (C-2022-055775, C-2023-047128).
Patent granted: KR 10-2980260; application KR 10-2024-0047575 [C2].
Supported by IITP grant funded by the Korea government (MSIT).

INDUSTRY EXPERIENCES

Data Mining Lab, KAIST AI

Mar. 2021 – Dec. 2021

Dynamic Stream Triangle Counting Algorithms. Advisor: Kijung Shin

Data Science Team, SK Hynix

Jun. 2020 – Aug. 2020

Discrimination of Defective Semiconductors using GNN

SETsystem

Jul. 2019 – Aug. 2019

Ship Detection: Differentiating True Lines from Noise using CNN

AWARDS & HONORS

KFAS (Korea Foundation for Advanced Studies) Scholarship

Field of Computer Science

2024 – 2029 (expected)

The PROUD POSTECHIAN MEMBER with the Highest Honor

Excellence Award (≈ \$400)

Feb. 2022

UGRP (Undergraduate Group Research Program)

Excellence Award (≈ \$4,000)

Jan. 2021

Presidential Science Scholarship by Republic of Korea

Full Tuition + Extra Fee (≈ \$8,000)

2020 – 2022

Jigok Scholarship by POSTECH

Full Tuition

2018 – 2020

PROFESSIONAL SERVICE & TEACHING

Reviewer AAAI 2026, AAAI 2026 (AI Alignment Track), KDD 2026 (AI for Science Track)

URP Teaching Assistant KAIST. Mentee: J.H. Lee → [C3]. Excellence Award. Feb. 2023 – Oct. 2023

Lab Mentor KAIST. Mentee: S.B. Yoon → [J1] Aug. 2022 – Jun. 2023

Teaching Mentor & Tutor POSTECH. Applied Linear Algebra, Analysis I, Programming and Problem Solving Fall 2019 – Fall 2021 (7 semesters)

Educational Volunteering Mentor POSTECH. Online Mentoring & Korea Scholarship Foundation Summer Camp Jul. 2019 – Dec. 2020

SKILLS

Programming Languages	C/C++, Python (DL: PyTorch), $\text{\LaTeX}$, Julia, R, JavaScript Korean (native), English (fluent: IELTS Overall 7.5; TOEFL – S 24, W 23)
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MISCELLANEOUS

Identifying Fraudulent Products in Secondhand Trading Websites

Mar. 2021 – Jun. 2021

POSTECH. IMEN472 Statistical Data Mining term project (1st place, model case). Implemented web crawlers, machine learning algorithms and performed statistical analysis.

Personality Prediction AI and App Development

Apr. 2020 – Jan. 2021

POSTECH. UGRP Excellence Award (Advisor: Jaesik Park). Developed and distributed an app platform based on a DL-based model with high accuracy.

Korean I-Corps Program: Team Lutra

Sep. 2021 – Oct. 2021

POSTECH. Korean I-Corps program for supporting preliminary startups. Title: Development of a Personalized Restaurant and Cafe Recommendation Web Service.

Winner of POSTECH-KAIST Science War*Sep. 2020**POSTECH. Science Quiz Session. Mathematics representative.***Leadership Experiences***Feb. 2020 – Dec. 2021**POSTECH. The student council of the Department of Mathematics (Head of Academic Affairs & Public Relations).**Orchestra club 'HanUIRim' (President & Conductor, Feb. 2020 – Dec. 2020).***Amateur Oboist & Classical Music Enthusiast***Guest player of several university orchestras (e.g., Sogang Univ., Sungkyunkwan Univ., Medical School Union, etc.).**Principal oboe player of Yeodo Elementary School Orchestra European tour for promoting International Exposition**Yeosu Korea 2012 (Venues: Musikverein, Mozarteum Grosser Saal, Slovak Radio Hall, World Trade Center Patria**Hall, Rudolfinum Dvorak Hall, Lukaskirche). 9th Online Supporters, i.e., certified influencers, of Seoul Philharmonic**Orchestra for promoting concerts.*